

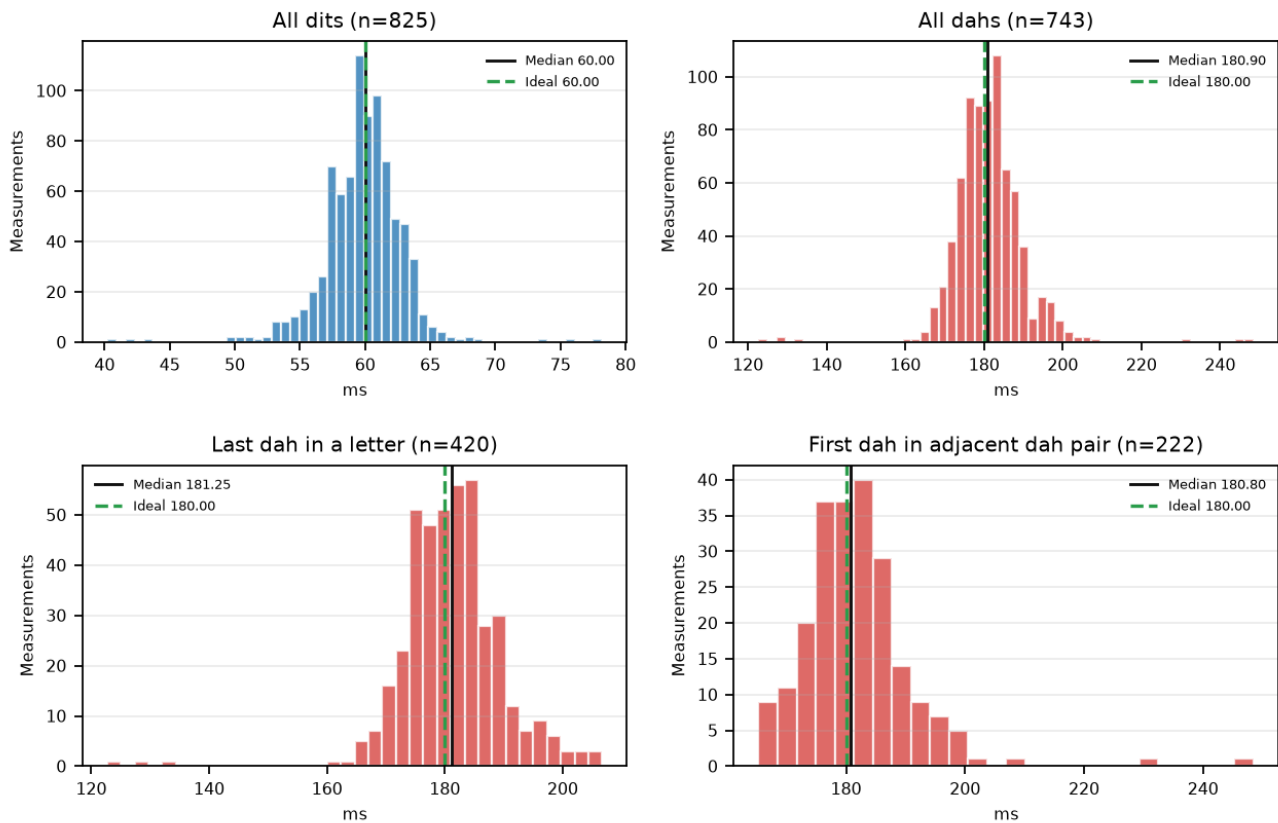
CW Analysis

CWAnalysis is produced by MOMZB

Source data: examples/sample_500_letters.csv

Accepted characters	500	Rejected records	0
Median dit	60 ms	Median dah	180.9 ms
Dash/dit ratio	3.015	Estimated speed	20.0 WPM
Fist quality score	90.2%	Rating	*****

Overall element distributions



Rejected input

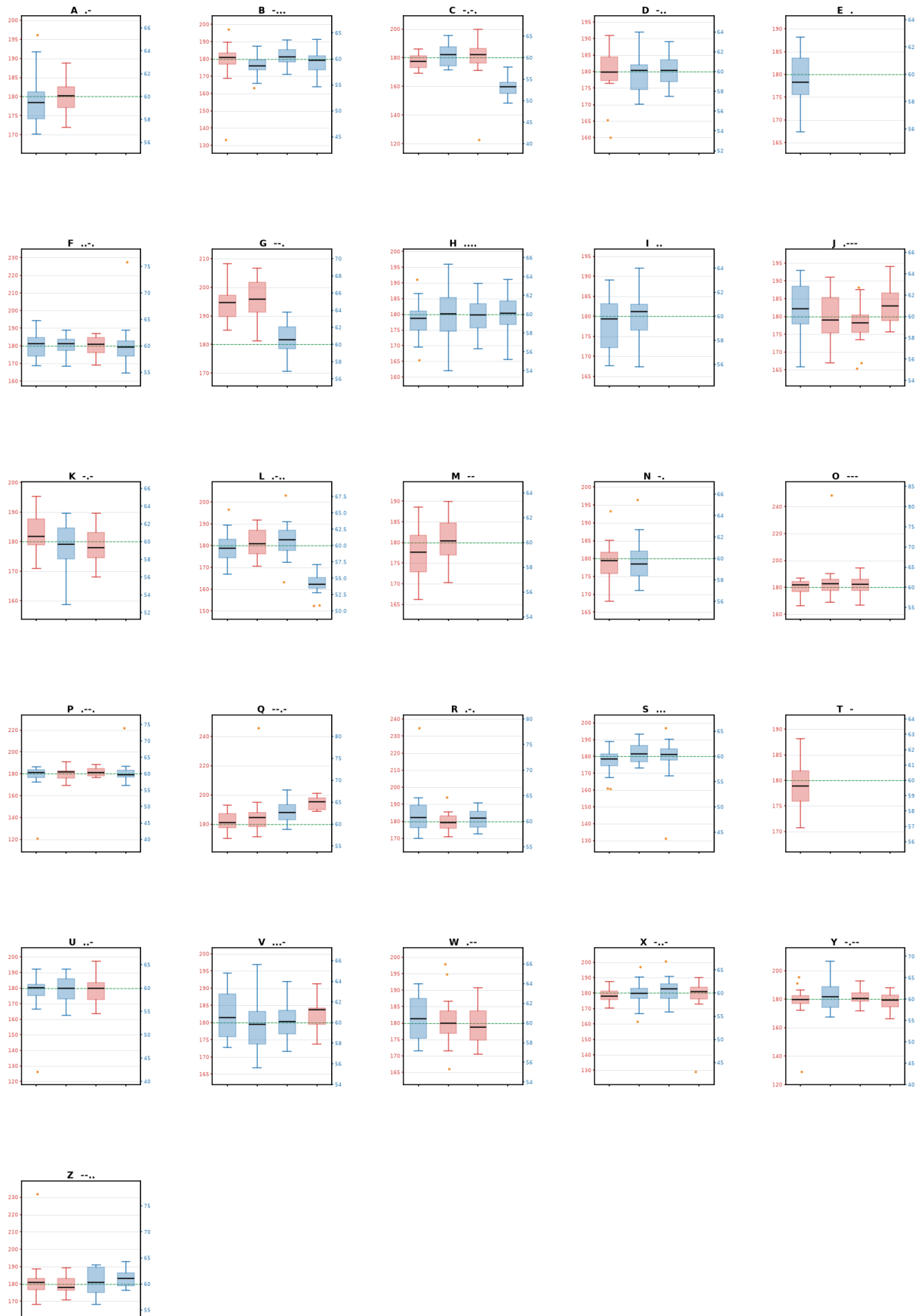
None

Recommendations

- Review position-dependent technique for: C, L, Q.

Letter Timing Summary

A-Z; each panel reserves four element positions. Dah timing is on the left and the linked 3:1 dit scale is on the right.



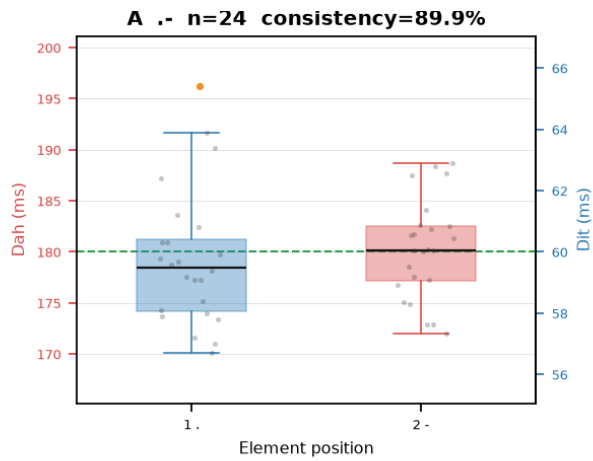
Number Timing Summary

0-9; each panel reserves all five element positions. Dah timing is on the left and the linked 3:1 dit scale is on the right.

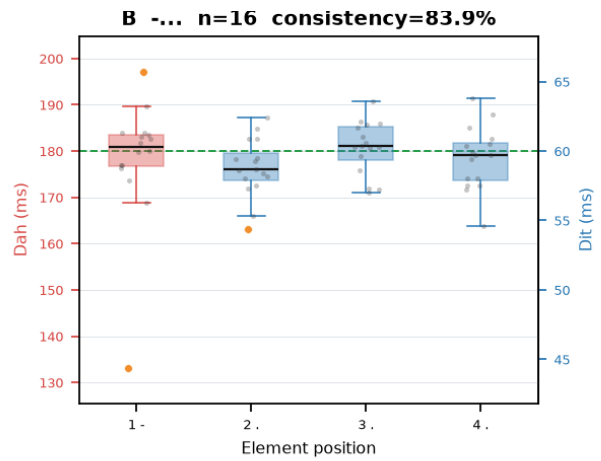


Detailed Character Analysis

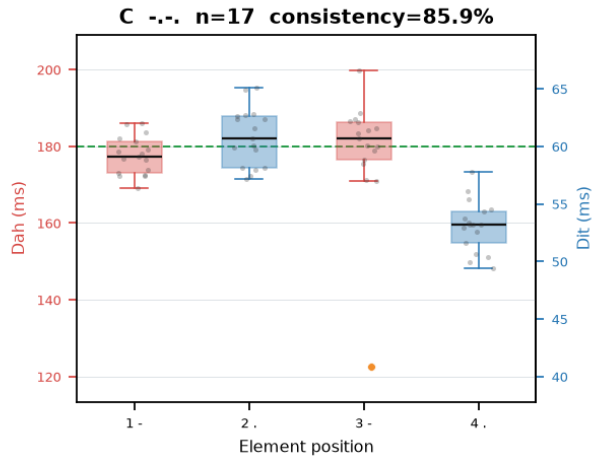
Source data: examples\sample_500_letters.csv



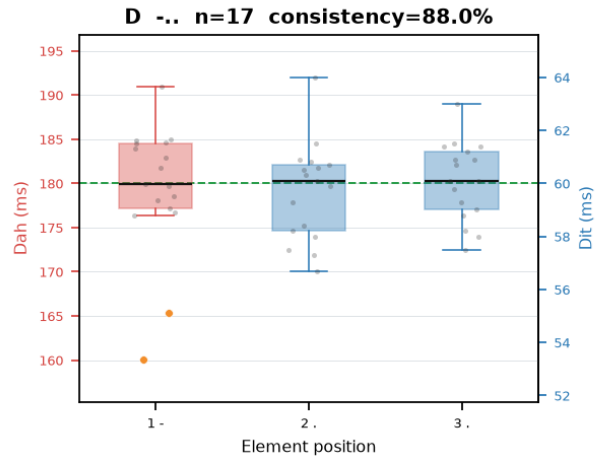
Best 2 | Worst 1 | No material position bias detected at the 5% threshold.



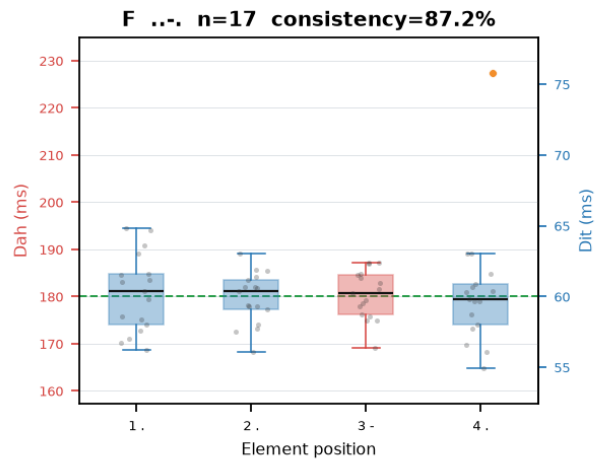
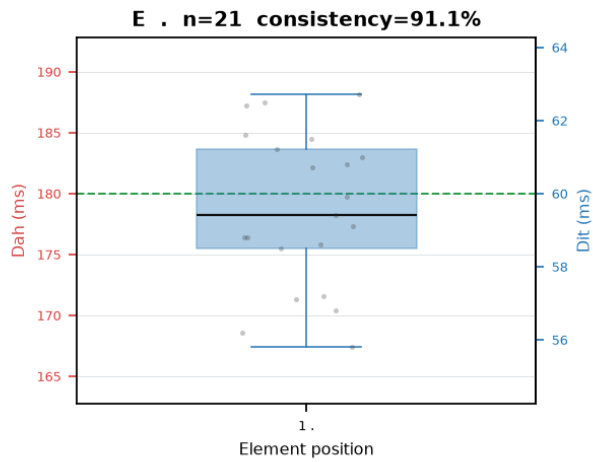
Best 3 | Worst 1 | No material position bias detected at the 5% threshold.



Best 1 | Worst 3 | Last dit is 12.4% shortened.

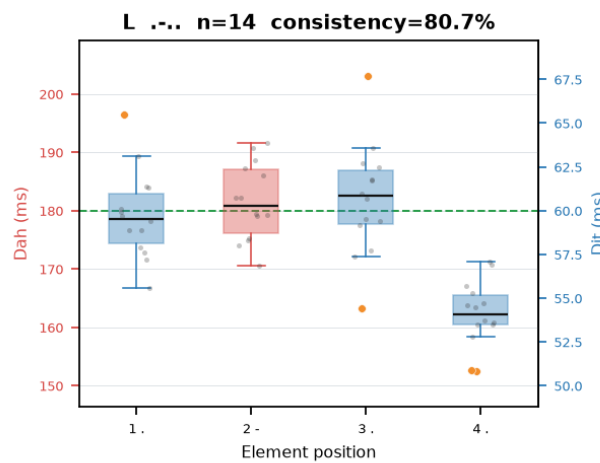
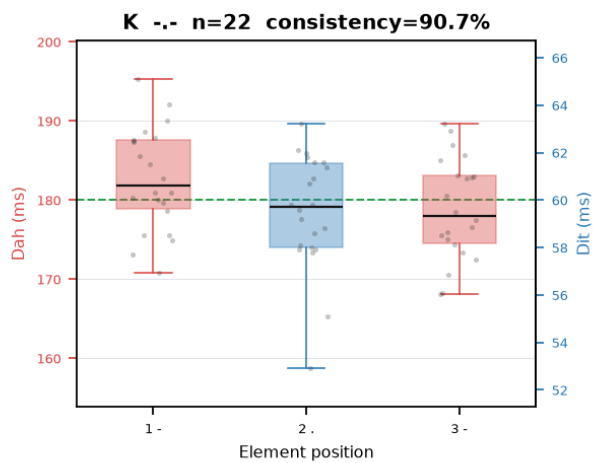
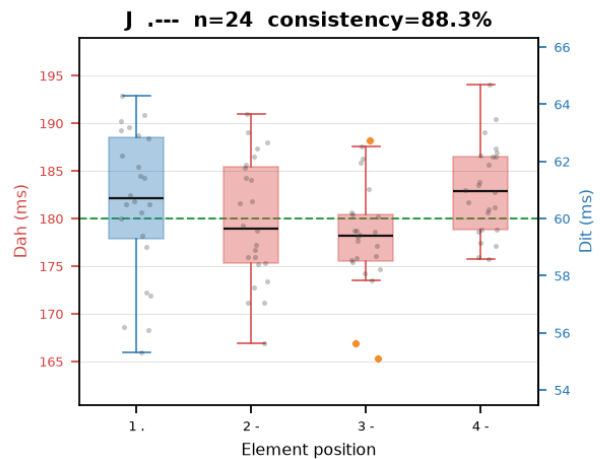
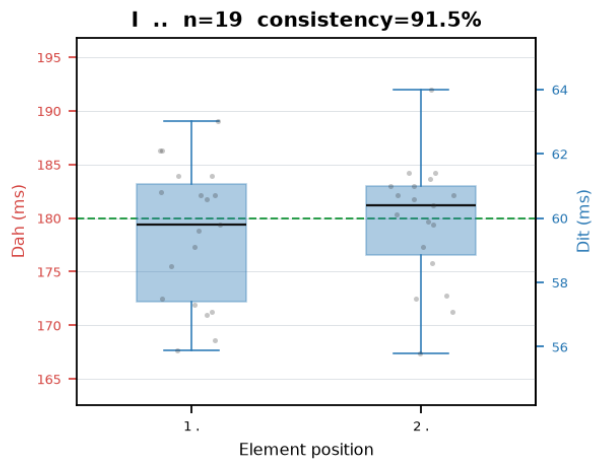
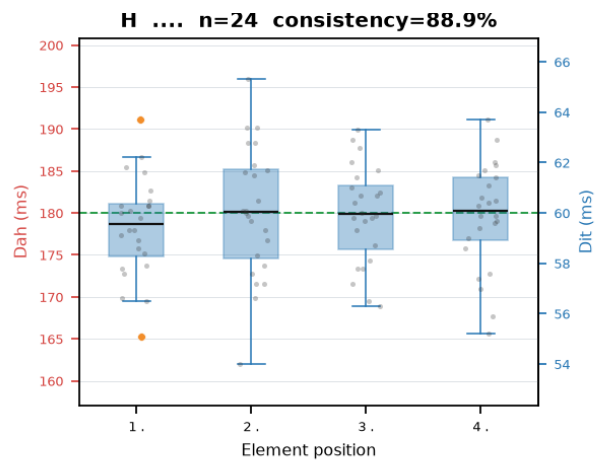
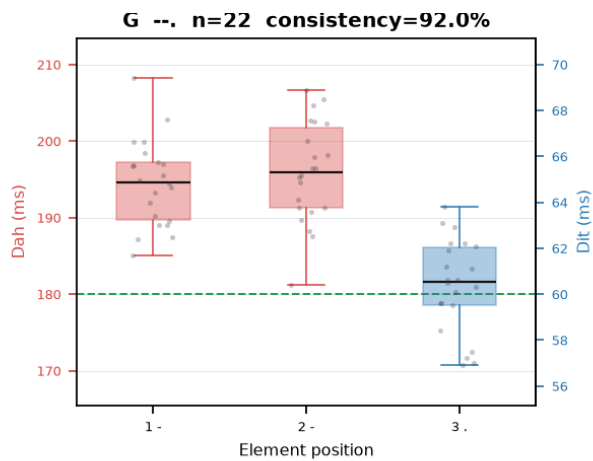


Best 3 | Worst 1 | No material position bias detected at the 5% threshold.



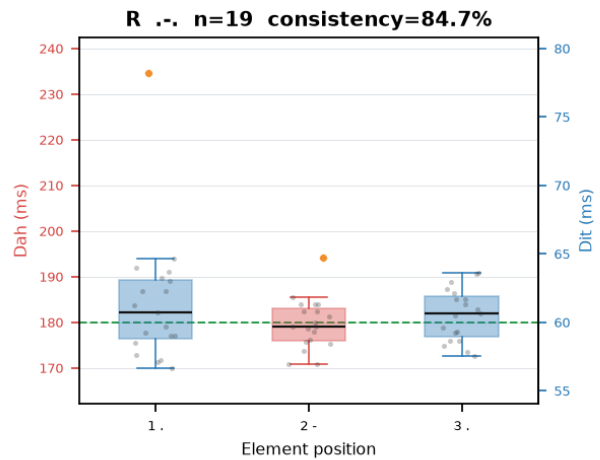
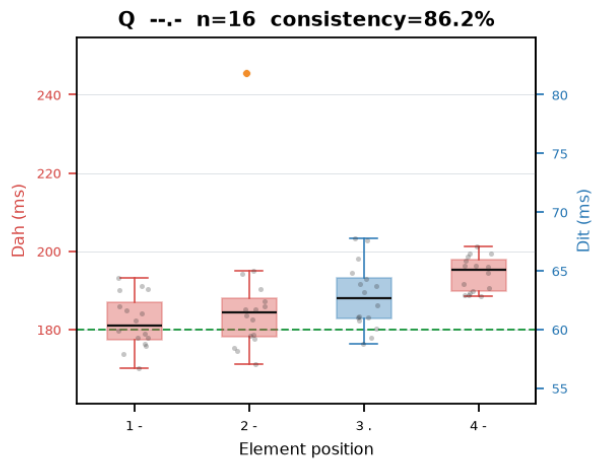
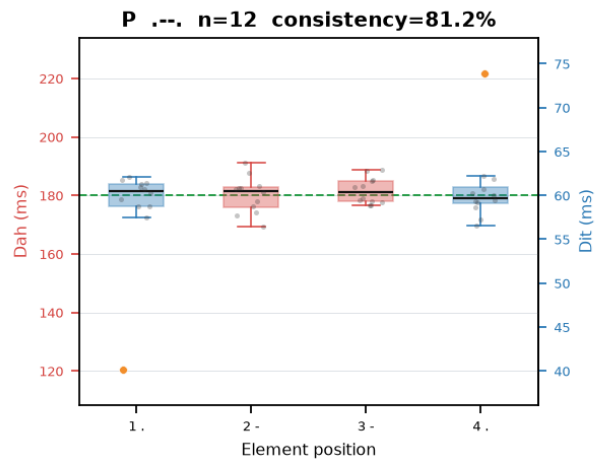
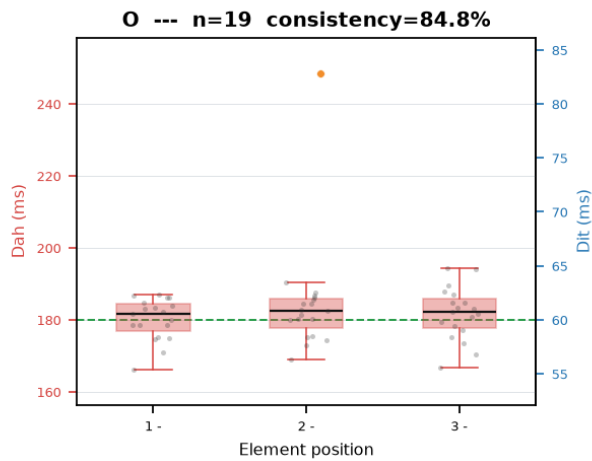
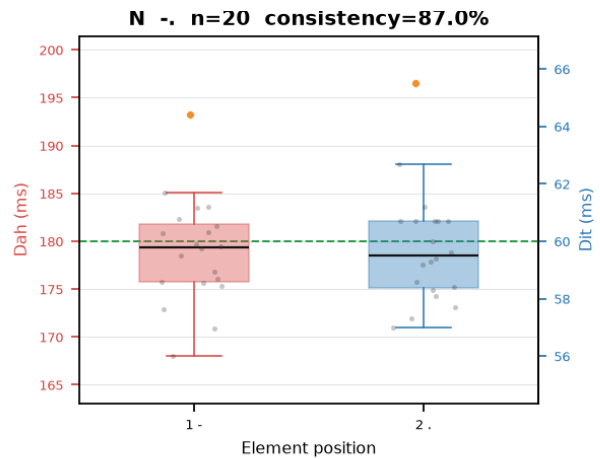
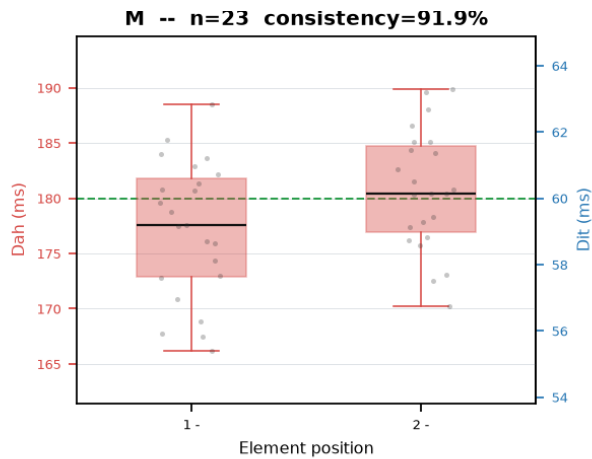
Detailed Character Analysis

Source data: examples\sample_500_letters.csv



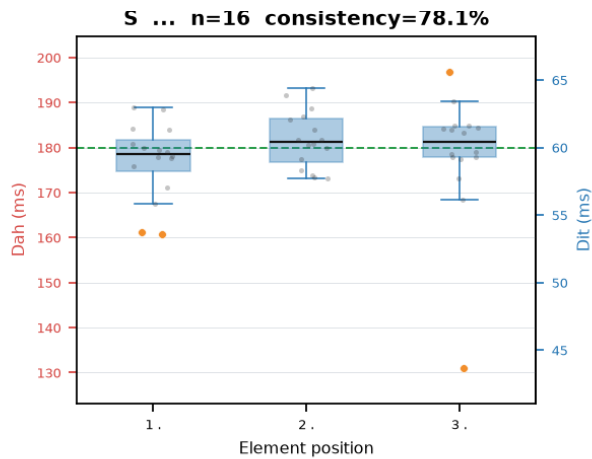
Detailed Character Analysis

Source data: examples\sample_500_letters.csv

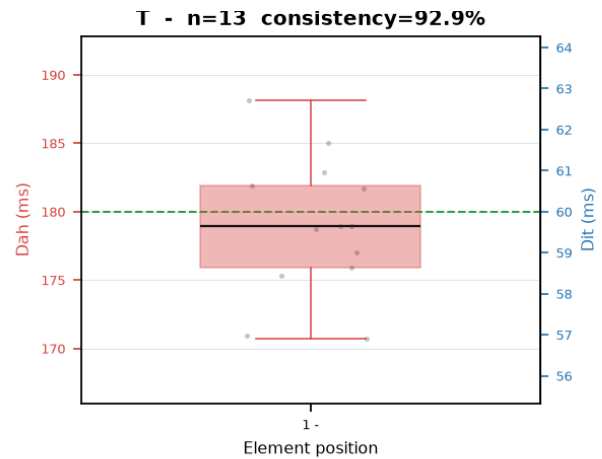


Detailed Character Analysis

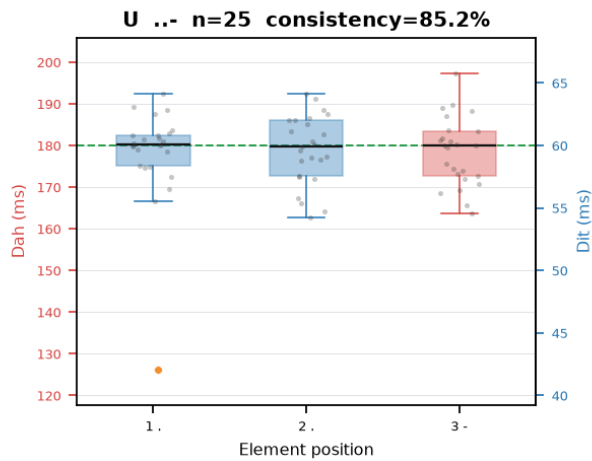
Source data: examples\sample_500_letters.csv



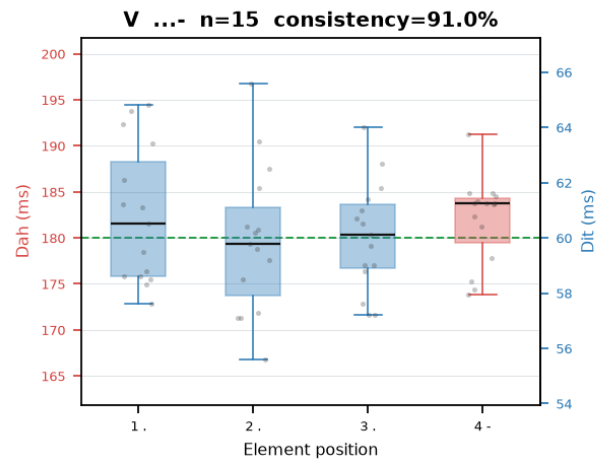
Best 2 | Worst 3 | No material position bias detected at the 5% threshold.



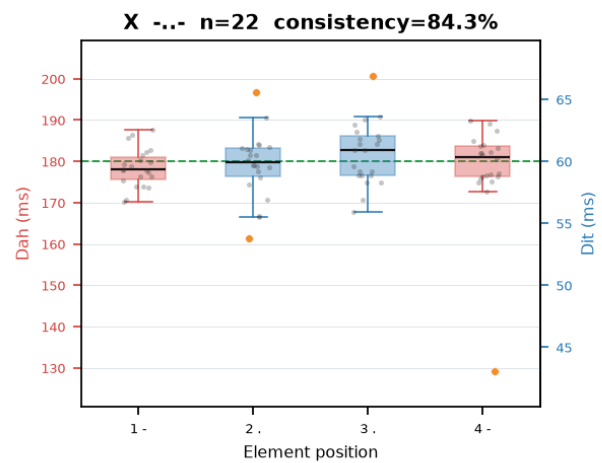
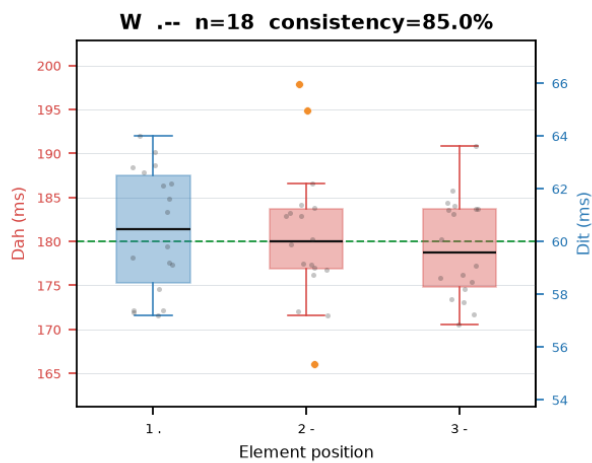
Best 1 | Worst 1 | Single-element character; position bias is not applicable.



Best 3 | Worst 1 | No material position bias detected at the 5% threshold.

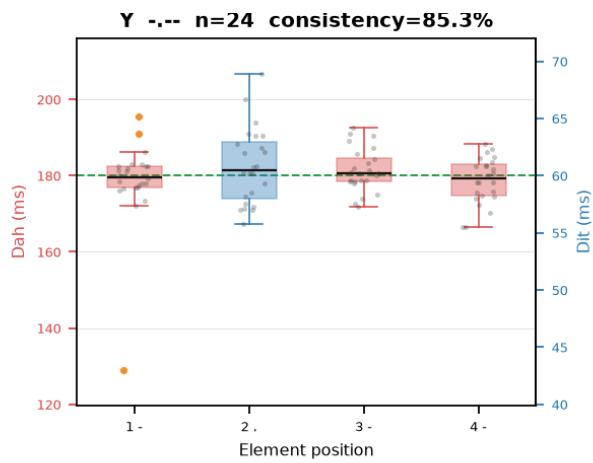


Best 4 | Worst 2 | No material position bias detected at the 5% threshold.

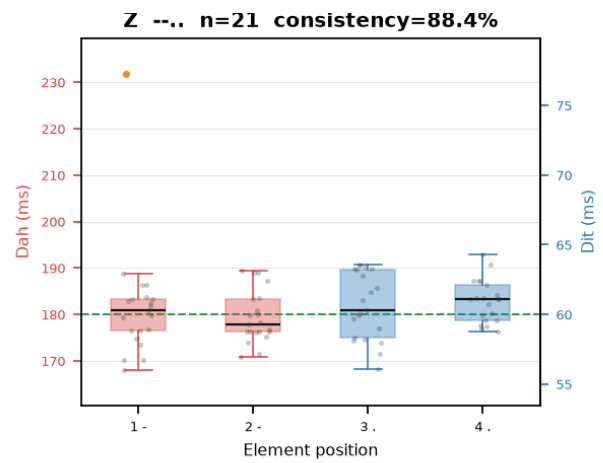


Detailed Character Analysis

Source data: examplesample_500_letters.csv



Best 3 | Worst 1 | No material position bias detected at the 5% threshold.



Best 4 | Worst 1 | No material position bias detected at the 5% threshold.

How to Read This Report

Axes and timing reference

Dah timing uses the red primary Y-axis on the left. Dit timing uses the blue secondary Y-axis on the right. The axes are linked at exactly 3:1: a dah at 180 ms is horizontally aligned with a dit at 60 ms. This permits direct visual comparison without making dits and dahs appear to use the same numerical duration.

The green dashed line marks the session median dit on the right axis and the corresponding ideal 3-times dah duration on the left. On the overview histograms, the black line is the observed median and the green line is the ideal timing reference.

The first page also isolates two dah populations across A-Z: the final dah occurring in each transmitted letter, and the first dah of every adjacent dah pair. Adjacent pairs overlap, so a three-dah run contributes positions 1 and 2 to the pair histogram.

Box-and-whisker plots

- The coloured box spans the 25th to 75th percentiles, also called the interquartile range (IQR).
- The black line inside the box is the median for that character and element position.
- The whiskers extend to the most extreme observations still within 1.5 IQR of the box.
- Small grey points are individual measurements. Their horizontal jitter only prevents overlap; it has no timing meaning.
- Orange points are outliers. With the default IQR method they fall below $Q1 - 1.5 \text{ IQR}$ or above $Q3 + 1.5 \text{ IQR}$.

Best, Worst and position bias

Best 2 means element position 2 has the lowest coefficient of variation (standard deviation divided by mean) within that character. Worst 1 means position 1 has the highest coefficient of variation. These labels describe repeatability, not whether the median duration is closest to ideal. With equal variability, the first tied position is reported.

A material first- or last-element bias is reported when its median differs by at least 5% from other elements of the same type in that character. For characters containing four or more elements, normalised timing drift is reported when the fitted change reaches 3% per position. Dahs are divided by three before that drift comparison.

No material position bias detected means none of those practical thresholds was crossed. It does not prove that the positions are statistically identical, especially when the sample count is small.

Scores and supporting statistics

The character consistency percentage combines average coefficient of variation and outlier rate. The overall fist-quality score also applies a penalty for departure from the ideal 3:1 median dah/dit ratio. Both are fixed, reproducible coaching indicators rather than calibrated scientific scores.

n is the number of accepted occurrences of the character. Complete per-position values - including mean, median, variance, standard deviation, MAD, quartiles, IQR, confidence interval, skewness, kurtosis and outlier rate - are exported to CW_Summary.csv beside this PDF.